

PROXIMITY PRIZE RESULTS

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ABSTRACT. We collect the strongest unconditional Reed–Solomon mutual correlated agreement results currently available in the four principal manuscripts and the theorem-ready part of the `rs-mca` repository. Exact affine-line staircases give complete thresholds for four maximal-dimension smooth prime-field rows, an exact $\mathbb{F}_{1732}$ row, additive binary rows, separating-field rows, and several saturated or post-Johnson examples. Unconditional corridor theorems give substantially larger safe radii at rates $1/4$, $1/8$, and $1/16$, while universal field-size caps and identity-prefix constructions provide complementary unsafe bounds near capacity. We also state the exact high-agreement line, curve, and interleaved-list ledger and summarize the strongest unconditional KoalaBear and Mersenne-31 structural packets. This is a results-only synthesis: proofs and certificate arithmetic are not reproduced, and every assertion is accompanied by a source where the proof or exact replay can be found.

1. SCOPE, CONVENTIONS, AND SOURCE HIERARCHY

The Proximity Prize asks, for a smooth Reed–Solomon code $C = \text{RS}_{\mathbb{F}}(D, k)$ and a target such as $\varepsilon^* = 2^{-128}$, for the largest radius at which the mutual correlated agreement error is at most ε^* [PP26, ABF26]. Here D has size n , usually a multiplicative coset of power-of-two order, and the official rates are $k/n \in \{1/2, 1/4, 1/8, 1/16\}$. This paper is a theorem catalogue, not a replacement for the cited proof papers.

For a nonempty challenge set $\Gamma \subseteq \mathbb{F}$, let $B_{C,\Gamma}^{\text{MCA}}(a)$ be the maximum number of slopes $\gamma \in \Gamma$ for which a point on a received affine line is explained by a degree- $< k$ polynomial on some support of size at least a , while the two generators of the received line are not simultaneously explained on that same support. Put

$$e_{\text{MCA},\Gamma}(C, \delta) = \frac{B_{C,\Gamma}^{\text{MCA}}(n - \lfloor \delta n \rfloor)}{|\Gamma|}, \quad B^* = \lfloor \varepsilon^* |\Gamma| \rfloor.$$

If integer radius r_0 is the first unsafe radius, the largest safe grid radius is $(r_0 - 1)/n$, while the supremum of the real safe set is r_0/n and is not attained.

The source hierarchy is as follows. Exact staircases, shortening bounds, finite rows, saturation statements, and paving compilers are in [ChoShort26, ChoThr26]. Universal field-size caps, smooth and circle transports, and deployed identity-prefix floors are in [ChoCap26]. The active structural and finite completion architecture is in [ChoGF26], while its companion reductions and $S+A+E$ handoff are in [ChoComp26]. External safe-side bounds are taken from [GKL24, Hab25] through the exact corridor audit [Lat26]. Repository-local theorems are cited to immutable notes or packets at commit `5ecb9ab5`; the coordination log is not used as proof authority.

2. EXACT AFFINE-LINE STAIRCASES

Theorem 2.1 (Support, tangent, and sparsification package). *Let $C = \text{RS}_{\mathbb{F}}(D, k)$, let $a \geq k+1$, put $r = n - a$, and let $\emptyset \neq \Gamma \subseteq \mathbb{F}$.*

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TABLE 1. Status vocabulary.

label	meaning
exact threshold	safe and unsafe sides meet at adjacent integer radii
proved safe edge	an unconditional safe radius without a matching adjacent unsafe theorem
proved unsafe edge	an unconditional failure radius without a matching adjacent safe theorem
bankable atom	an exact contribution inside a fixed first-match partition
local theorem / route cut	an unconditional implication under its printed branch contract, but not an exhaustive row theorem
conditional packet	a proved compiler whose named pointwise or global hypothesis is still open

- (i) *Every bad slope has a witness on a support of size exactly a , and one exact support carries at most one bad slope. Hence*

$$B_{C,\Gamma}^{\text{MCA}}(a) \leq \min \left\{ |\Gamma|, \binom{n}{a} \right\}.$$

- (ii) *The universal tangent construction gives*

$$B_{C,\Gamma}^{\text{MCA}}(a) \geq \min\{|\Gamma|, r + 1\}.$$

- (iii) *If $B_{C,\Gamma}^{\text{CA}}(a)$ is the column-far correlated-agreement numerator and $\sigma_{C,\Gamma}(a)$ the sparse mutual numerator with union support at most r , then*

$$B_{C,\Gamma}^{\text{MCA}}(a) = \max\{B_{C,\Gamma}^{\text{CA}}(a), \sigma_{C,\Gamma}(a)\}.$$

- (iv) *If $2r \leq n - k$, then $\sigma_{C,\Gamma}(n - r) = \min\{|\Gamma|, r\}$.*

Source and status. [ChoShort26, ChoThr26]; the exact support-wise line formulation is also in [Line26].

Theorem 2.2 (Quadratic staircase). *If $0 \leq r \leq n - k - 1$ and*

$$(n - r)^2 \geq n(k + r),$$

then

$$B_{C,\Gamma}^{\text{MCA}}(n - r) = \min\{|\Gamma|, r + 1\}.$$

Equivalently, the formula holds for every $0 \leq r \leq r_{\text{quad}}(n, k)$, where

$$r_{\text{quad}}(n, k) = \left\lfloor \frac{3n - \sqrt{n(5n + 4k)}}{2} \right\rfloor.$$

Source and status. [ChoShort26, Quadratic mean-overlap theorem] and [ChoThr26, Quadratic staircase].

Corollary 2.3 (Exact target compiler). *Let $B = \lfloor \varepsilon |\Gamma| \rfloor$ and assume*

$$1 \leq B < |\Gamma|, \quad B \leq n - k - 1, \quad (n - B + 1)^2 \geq n(k + B - 1).$$

Then

$$\{\delta : e_{\text{MCA},\Gamma}(C, \delta) \leq \varepsilon\} = [0, B/n],$$

and the endpoint B/n is unsafe.

Source and status. [ChoShort26, ChoThr26, Exact target compiler].

Theorem 2.4 (Official-rate staircase almost to half distance). *Assume $n = 2^e \geq 128$, $k = \rho n$, and $\rho \in \{1/2, 1/4, 1/8, 1/16\}$. Put*

$$r_{\rho}^{\#} = \begin{cases} n/4 - 3, & \rho = 1/2, \\ 3n/8 - 2, & \rho = 1/4, \\ 7n/16 - 2, & \rho = 1/8, \\ 15n/32 - 2, & \rho = 1/16. \end{cases}$$

For every nonempty Γ and every $0 \leq r \leq r_\rho^\sharp$,

$$B_{C,\Gamma}^{\text{MCA}}(n-r) = \min\{|\Gamma|, r+1\}.$$

Whenever $1 \leq B = \lfloor \varepsilon |\Gamma| \rfloor < |\Gamma|$ and $B \leq r_\rho^\sharp + 1$, the complete safe set is $[0, B/n)$.

Source and status. [ChoShort26, ChoThr26]; the CA input is the two-radius theorem of [BCHKS25].

3. COMPLETE SOLVED ROWS

Theorem 3.1 (Four maximal-dimension smooth prime-field thresholds). *For every row in Table 2, p is an explicit Proth prime below 2^{256} , $n \mid p-1$, the subgroup $D \leq \mathbb{F}_p^*$ of order n exists, $k = 2^{40}$, and at target 2^{-128} the complete safe set is $[0, B/n)$, where $B = \lfloor p/2^{128} \rfloor$. Every endpoint is unsafe.*

TABLE 2. Complete smooth-domain thresholds at maximal permitted message dimension.

rate	n	B	explicit prime p
1/2	2^{41}	389500552609	13254016995880403333249306710494641010898987122689
1/4	2^{42}	1210584858040	411940680852499481698306614369841346700408394874881
1/8	2^{43}	2879806199253	979947269755402568812854322316630667196565607677953
1/16	2^{44}	6233898019554	2121285573237585848299875619011192262679065433997313

Source and status. [ChoShort26, Reproducible finite Prize instances] and [ChoThr26, Four certified smooth rows].

Theorem 3.2 (Exact $\mathbb{F}_{17^{32}}$ smooth threshold). *Let $H \leq \mathbb{F}_{17^{32}}^*$ have order 512 and let $C = \text{RS}_{\mathbb{F}_{17^{32}}}(H, 256)$. At target 2^{-128} ,*

$$B_C^{\text{MCA}}(506) = 7, \quad B_C^{\text{MCA}}(507) = 6, \quad \lfloor 17^{32}/2^{128} \rfloor = 6.$$

Hence the complete real safe set is

$$[0, 6/512),$$

the largest safe grid radius is $5/512$, and $6/512 = 3/256$ is the unsafe supremal endpoint.

Source and status. The tangent staircase and tangent-star exclusion are in [F17-26]; the prize-gate dependence is independently audited in [F17-A].

Corollary 3.3 (Binary additive-domain thresholds). *Let $k = 2^{40}$ and let D be an $\mathbb{F}_2$ -linear subspace of the indicated field with the indicated size. At target 2^{-128} , the complete safe sets are*

ρ	$\mathbb{F}$	n	safe set
1/2	$\mathbb{F}_{2^{166}}$	2^{41}	$[0, 1/8)$
1/4	$\mathbb{F}_{2^{168}}$	2^{42}	$[0, 1/4)$
1/8	$\mathbb{F}_{2^{169}}$	2^{43}	$[0, 1/4)$
1/16	$\mathbb{F}_{2^{170}}$	2^{44}	$[0, 1/4)$.

Every endpoint is unsafe.

Source and status. [ChoShort26, ChoThr26, Binary additive-domain rows]. These are additive FFT domains rather than multiplicative smooth domains.

Theorem 3.4 (Saturated and post-Johnson finite rows). *Let $p = (2^{128} - 255)2^{128} + 1$. At target 2^{-128} ,*

- (i) if $D \leq \mathbb{F}_p^*$ has order 128 and $k \in \{64, 32, 16, 8\}$, the complete safe set is $[0, 1]$;

(ii) if $D \leq \mathbb{F}_p^*$ has order 512, the rows

$$(k, a) = (256, 351), (128, 233), (64, 143), (32, 52)$$

are safe at radii 161/512, 279/512, 369/512, and 115/128, respectively, and every point is strictly beyond the exact MDS Johnson radius.

Source and status. Official saturation and the four length-512 shortening/circuit certificates are in [ChoShort26].

Theorem 3.5 (First adjacent threshold over separating fields). Put $M = \binom{n}{k+1} = \binom{n}{n-k-1}$. If

$$|\mathbb{F}| > \max \left\{ M, \binom{M}{2} \right\},$$

then $B_C^{\text{MCA}}(k+1) = M$. For $b = \lfloor \varepsilon |\mathbb{F}| \rfloor$, the first safe agreement is $k+1$ when $b \geq M$, and it is $k+2$ whenever

$$\binom{n}{k+2} \leq b < M.$$

Source and status. [ChoGF26, Exact first adjacent row over a separating field].

Theorem 3.6 (Large scalar-extension capacity regime). Let $C_n = \text{RS}_{\mathbb{F}_n}(D_n, k_n)$ with $D_n \subseteq \mathbb{B}_n \subseteq \mathbb{F}_n$, $|D_n| = n$, and $k_n/n \rightarrow \rho$. If

$$\frac{a_n}{n} \rightarrow \alpha, \quad (a_n - k_n - 1) \log |\mathbb{B}_n| = o(n),$$

and

$$\frac{1}{n} \log |\mathbb{F}_n| \rightarrow \phi, \quad \frac{1}{n} \log (|\mathbb{F}_n| - n) \rightarrow \phi,$$

then $\alpha = \rho$ and

$$\frac{1}{n} \log B_{C_n}^{\text{MCA}}(a_n) = \min\{h(\rho), \phi\} + o(1).$$

For targets $\varepsilon_n = e^{-\lambda n + o(n)}$ with $\lambda < \phi - h(\rho)$, the first permitted agreement $k_n + 1$ is safe and the threshold is $1 - \rho + o(1)$.

Source and status. [ChoGF26, Large scalar-extension exponent].

4. STRONG UNCONDITIONAL SAFE-SIDE BOUNDS

Theorem 4.1 (Audited corridor safe edges). The unconditional MCA bounds of Gao–Kan–Li and Haböck, instantiated with the exact challenge budgets in [Lat26], certify the safe radii in Table 3. The Haböck bound is the stronger of the two in all six rows.

TABLE 3. Audited unconditional corridor safe edges at target 2^{-128} .

row	n	safe integer radius	safe relative radius
length-1024, rate 1/4	2^{10}	512	0.500000
length-1024, rate 1/8	2^{10}	663	0.647461
length-1024, rate 1/16	2^{10}	769	0.750977
prize scale, rate 1/4	2^{41}	1092724518963	0.496914...
prize scale, rate 1/8	2^{41}	1415997755216	0.643921...
prize scale, rate 1/16	2^{41}	1644686143216	0.747917...

Source and status. The underlying theorems are [GKL24, Theorem 3] and [Hab25, Theorem 2]. The exact row arithmetic, adjacent failure checks, and provenance audit are in [Lat26].

Remark 4.2. The prize-scale corridor packet uses a pinned exact budget convention corresponding to a line field near $2^{255.9}$; it does not yet pin one literal prime in the paper. The safe inequalities are unconditional under that stated denominator. A final row release should add a literal admissible field and rerun the same exact comparisons.

Theorem 4.3 (Generalized exact high-agreement ledger). *Let $C = \text{RS}_{\mathbb{F}}(D, k)$, put $R = n - k$, and write $r = n - a$.*

- (i) *If $r \leq \lfloor R/3 \rfloor$, then the finite affine support-wise MCA numerator, no-loss CA numerator, and projective-slope support-wise numerator all equal $r + 1$.*
- (ii) *For every interleaving arity $\mu \geq 1$, if $r \leq \lfloor R/2 \rfloor$, then the common-support list size satisfies $\Lambda_{\mu}(C, a) = 1$.*
- (iii) *For the finite-parameter degree- d power curve*

$$f_0 + \gamma f_1 + \cdots + \gamma^d f_d,$$

if $r \leq \lfloor R/(d+2) \rfloor$, then the support-wise curve-MCA and no-loss curve-CA numerators both equal $\min\{|\mathbb{F}|, d(r+1)\}$.

- (iv) *If a coding ledger uses curve or line degrees $d_1, \dots, d_s$ and ℓ interleaved-list terms over one denominator Q , then in the common exact range*

$$N_{\text{coding}}(r) = \ell + \sum_{i=1}^s d_i(r+1).$$

Source and status. [High26]. The special $\mathbb{F}_{17^{32}}$ adjacent line/list ledger is in [F17-26].

Remark 4.4 (Line-decoding terminology). The support-wise numerator LD_{sw} satisfies

$$e_{\text{MCA}}(C, \delta) = \frac{LD_{\text{sw}}(C, \lceil (1-\delta)n \rceil)}{|\mathbb{F}|}.$$

This is not the stronger Goyal–Guruswami notion of (δ, A, B) line-decodability. The latter implies MCA through Theorem 4.21 of [ABF26], whereas ordinary close-point line decoding can be strictly larger than the support-wise numerator. The exact bridge, residue-line reformulation, and separation example are in [Line26].

Corollary 4.5 (List consequences of CA/MCA bounds). *Let $C = \text{RS}_{\mathbb{F}}(D, k)$ and $C^+ = \text{RS}_{\mathbb{F}}(D, k+1)$.*

- (i) *Under the hypotheses of Theorem 5.2 of [ABF26], sufficiently small CA error implies $|\Lambda(C, \delta)| < |\mathbb{F}|$.*
- (ii) *Under the hypotheses of Theorem 5.3 of [ABF26], a CA error bound for C gives*

$$|\Lambda(C^+, \delta)| \leq \left\lceil \frac{|\mathbb{F}|}{1-\eta} e_{\text{CA}}(C, \delta) \right\rceil.$$

- (iii) *In the quadratic staircase range, substituting the exact MCA numerator $r + 1$ gives the explicit integer consequences stated in [ChoThr26, Line decoding and list-decoding consequences].*

Source and status. Theorems 5.2 and 5.3 of [ABF26]; the exact deep-point form of item (ii) is also in [ChoCap26].

5. UNIVERSAL UNSAFE CAPS AND EXACT FAILURE EDGES

Theorem 5.1 (Universal field-size cap). *Let $\rho \in \{1/2, 1/4, 1/8, 1/16\}$, let $|\mathbb{F}| < 2^{256}$, let $D \subseteq \mathbb{B}^*$ be a multiplicative coset of order n , and let $k = \rho n \leq 2^{40}$. Assume $1024 \mid n$ for $\rho \geq 1/8$ and $2048 \mid n$ for $\rho = 1/16$. At target 2^{-128} ,*

$$\delta_C^* \leq 1 - \rho - 2^{-9} \quad (\rho = 1/2, 1/4, 1/8),$$

and

$$\delta_C^* \leq 1 - \rho - 2^{-10} \quad (\rho = 1/16).$$

Throughout the corresponding terminal interval the MCA error is greater than 2^{-86} ; if $|\mathbb{F}| \geq 2n$, it is greater than 2^{-42} .

Source and status. [ChoCap26, Universal field-size cap].

Corollary 5.2 (Map-smooth and circle transport). *The same cap applies, with the multiplicity correction stated in [ChoCap26], to map-smooth images whose fibers have uniformly bounded size. In particular, it applies to the normalized circle/Chebyshev presentation after accounting for the two-to-one torus map and the exceptional branch points.*

Source and status. [ChoCap26, Map-smooth and circle cap theorems].

Theorem 5.3 (Exact deployed identity-prefix failure edges). *For the deployed rows with $n = 2^{21}$ and $k = 2^{20}$, the following agreements are unconditionally unsafe:*

row	target	a_0	unsafe radius $(n - a_0)/n$
KoalaBear sextic MCA	2^{-128}	1116047	981105/2097152
KoalaBear sextic list	2^{-128}	1116046	490553/1048576
Mersenne-31 circle MCA	2^{-100}	1116023	981129/2097152
Mersenne-31 circle list	2^{-100}	1116022	490565/1048576.

The witness families are identity-prefix locator fibers; at the printed odd agreements on a power-of-two domain, the supports are aperiodic.

Source and status. [ChoCap26, Corrected deployed identity-prefix floors]; the exact frontier manifests are in [Front26].

TABLE 4. Best current threshold statements for selected Prize-facing rows.

row	proved safe through	proved unsafe from	status
four Proth rows of Table 2	$(B - 1)/n$	B/n	exact adjacent threshold
$\mathbb{F}_{17^{32}}$, $n = 512$, $k = 256$	5/512	6/512	exact adjacent threshold
prize corridor, rate 1/4	1092724518963/ 2^{41}	383/512	unconditional interval
prize corridor, rate 1/8	1415997755216/ 2^{41}	447/512	unconditional interval
prize corridor, rate 1/16	1644686143216/ 2^{41}	959/1024	unconditional interval
KoalaBear sextic MCA, 2^{-128}	1/4 with BCIKS	981105/ 2^{21}	deployed interval; adjacent safe row open
Mersenne-31 circle MCA, 2^{-100}	1/4 with BCIKS	981129/ 2^{21}	stress-row interval; adjacent safe row open

Source and status. Safe inputs: [BCIKS20, Lat26]. Unsafe inputs: [ChoCap26]. Exact rows: [ChoShort26, ChoThr26, F17-26].

6. PRIMARY STRUCTURAL RESULTS AT THE DEPLOYED FRONTIER

The next statements are unconditional theorems used by the current finite completion architecture. They are not, individually or collectively, advertised here as a completed adjacent-row proof.

Theorem 6.1 (Saturation and split-pencil payments). *Let U be a received word and let $s_c(U)$ be its agreement count with $c \in C$.*

(i) *The exact support-locator saturation count at agreement m is*

$$\text{Cen}(U; m) = \sum_{c \in C} \binom{s_c(U)}{m}.$$

Thus raw support multiplicity must not be identified with codeword or slope multiplicity.

- (ii) For a projective polynomial pencil $sA + tB$, with common domain-root locator of degree g , if every counted member has at least h moving roots outside the common root set, then the number of counted projective parameters is at most

$$\left\lfloor \frac{n-g}{h} \right\rfloor.$$

- (iii) After the tangent, common-support, quotient, extension, degree-drop, and common-GCD branches are removed, every primitive one-parameter split-pencil chart at the active KoalaBear or Mersenne MCA row contributes at most two finite bad slopes.

Source and status. [ChoGF26, Saturation identity and moving-root split-pencil theorem].

Theorem 6.2 (Active support-flag and paving payments). *For the first unpaid affine Q /list charts, the following row-sharp payments are unconditional.*

- (i) Every support-closure codimension at least two is paid by the all-extender support-flag certificate when

$$b_0 \geq 70230 \quad \text{on KoalaBear}, \quad b_0 \geq 108962 \quad \text{on Mersenne-31}.$$

At the transition, the certified maxima are

row	certified maximum
KoalaBear MCA	274974976450914526
KoalaBear list	274975238687487221
Mersenne-31 MCA	16776934
Mersenne-31 list	16776950.

- (ii) The exact paving-basis defect thresholds are

KoalaBear nullity	0–9	10	11	12	13
$J_{B^*}^{\text{pav}}$	67471	67468	67423	66715	55724

and

Mersenne nullity	0–1	2	3	4	5
$J_{B^*}^{\text{pav}}$	67447	67432	67213	63797	10700.

Source and status. [ChoGF26, Active higher-codimension flag payment and exact paving-basis MCA payments]; the paving basis is [ChoShort26].

Theorem 6.3 (First Mersenne primitive cell and the Delsarte barrier). *The following statements are unconditional.*

- (i) For the Mersenne MCA constant-weight cell

$$N = 1053557, \quad h = 72435, \quad |X \cap Y| \leq 4981,$$

one has

$$|\mathcal{C}| \leq 16760701 < 16777216.$$

Thus the first Mersenne MCA primitive chart with nullity 4982, minimum-support excess 1, and common outside mismatch $b_0 = 6$ is paid.

- (ii) For the associated auxiliary-list parameters $N = 1053558$, $h = 72435$, and minimum Johnson distance 67454, the complete ordinary Johnson-scheme Delsarte LP optimum is

$$16838532.3143506 \dots > 16777215.$$

No ordinary Bose–Mesner/Delsarte polynomial of any degree can close that atom.

- (iii) A terminal counterexample of size 16387300 would contain at least 130003626982983 minimum-distance pairs, at least 127368440 edge-disjoint and 3749 vertex-disjoint K_{32} critical cores, and at least 1133314 projective minimum directions.
- (iv) Every coherent rational critical core has canonical owner size at least 440837. Among the edge-disjoint cores, one branch type occurs at least 63684220 times; in the rational branch some fixed eight-coordinate set lies in at least 59838 canonical owners.

Source and status. The cubic certificate, exact full Delsarte optimum, critical-core abundance, field descent, and owner concentration are in [ChoComp26].

7. COLLABORATIVE KOALABEAR AND MERSENNE THEOREM BANK

KoalaBear.

Theorem 7.1 (Source-coordinate tangent atom). *At the KoalaBear MCA agreement $a = 1116048$, every received line has a source-coordinate tangent first-match cell containing at most*

$$\boxed{981104}$$

distinct bad finite slopes. In the frozen four-cell chronology

$$Z = Z_{\text{paid}} \sqcup Z_Q \sqcup Z_{\text{BC}} \sqcup Z_{\text{new}},$$

this gives $U_{\text{paid}} = 981104$ and exact remaining reserve

$$274980728110413983.$$

Source and status. Holm Buar, [Bua26a]. The theorem pays only the tangent cell; U_Q , U_{BC} , and U_{new} remain open.

Theorem 7.2 (Base-field coordinate-span confinement). *Let $\mathbb{B} = \mathbb{F}_p \subseteq \mathbb{F}$, let $D \subseteq \mathbb{B}$, and let y_0, y_1 be the syndromes of a received line. Every nonzero finite bad slope satisfies*

$$\gamma \in L_{\mathbb{B}}(y_0)L_{\mathbb{B}}(y_1)^{-1}.$$

If $s = \dim_{\mathbb{B}} L_{\mathbb{B}}(y_0)$ and $u = \dim_{\mathbb{B}} L_{\mathbb{B}}(y_1)$, then the affine bad set has size at most

$$1 + \frac{(p^s - 1)(p^u - 1)}{p - 1},$$

and the projective bad set has size at most the same quantity plus one. A dependent syndrome pair has one bad projective parameter, and a syndrome plane containing two rank-one projective points has projective bad set of size at most $p + 1$.

Source and status. Danny, [Dan26b]. The exact rank-(1, 2) radial occupancy problem remains open.

Theorem 7.3 (Pole-tolerant scalar-locator localization). *For a coherent scalar-locator certificate*

$$Qh_i + (c_0 + c_1\gamma_i)\Lambda_i = A + \gamma_i B,$$

define

$$G_Q = \{x \in D : A(x) = Q(x)r_0(x), B(x) = Q(x)r_1(x)\}, \quad g = |G_Q|.$$

Every coordinate outside G_Q lies in at most one selected support. Consequently,

$$|I| \leq \left\lfloor \frac{n - g}{m - g} \right\rfloor \leq n - m + 1 \quad (g < m),$$

and

$$|I| \leq n - g + B_{\mathbb{F}, G_Q, k, \Gamma}^{\text{MCA}}(m) \quad (g \geq m).$$

For exact monic locators, after deleting at most one zero-scalar slope, every common denominator pole is a common support point and can be divided from Q, A, B , and all live locators while preserving the scalar-locator identity and degree profile.

Source and status. Danny, [Dan26a]. This is a local algebraic theorem; a chronology-correct first-match compiler is still required.

Theorem 7.4 (KoalaBear rank-nine moving-root boundary cut). *Every qualifying post-restart full-outside rank-two record surviving the source-rational owner satisfies*

$$s \geq 67474, \quad e \geq 33737, \quad \deg \gcd(P, Q) \leq 1014838.$$

The zero-source-slack boundary is absorbed by the earlier projective-base C5 owner. The first surviving one-slack components are the nonbase common-linear-GCD twist and the split-GCD nonbase moving-cofactor component.

Source and status. Scott Hughes, [Hug26a]. This is a route cut with zero row-level ledger movement.

Mersenne-31.

Theorem 7.5 (Source adapter and rank-46 terminal). *For the Mersenne-31 list row at agreement 1116023 and budget 16777215:*

- (i) *the exact low-weight cell contributes at most 3730 actual target-field codewords;*
- (ii) *the complete target-field list has a legal low/boundary/interior exact-support partition;*
- (iii) *every forbidden list forces at least 259881 marked rank-46 diagnostic keys;*
- (iv) *every marked key carries a coupled three-row locator–numerator frame with partial-degree bounds*

$$20765, \quad 41530, \quad 62295 < 67447.$$

Source and status. Scott Hughes, [Hug26b]. The marked keys are diagnostics, not charged cells; the boundary/interior residual remains unpaid.

Theorem 7.6 (All-weight boundary and fixed- G reductions). *For the same row:*

- (i) *every nonanchor codeword is in bijection with a canonical divisor/GCD pair (G, b) in the actual-anchor module, and fresh-symbol forcing reduces every live violation to a boundary-anchor census with arbitrary unit V ;*
- (ii) *for fewer than $p - 1$ reduced pairs, one common unit V realizes all declared exact full GCDs if and only if the individual coprimality gates and all pairwise Wronskian intersection/symmetric-difference gates hold;*
- (iii) *every base-field ordinary Reed–Solomon list on an R -point subset of the deployed domain embeds into one fixed- G Mersenne boundary slice; therefore row safety forces*

$$M_{\text{ord}} \leq 16777214.$$

Source and status. Scott Hughes, [Hug26c, Hug26g, Hug26d].

Theorem 7.7 (Affine-rank closures). *For every reconstructed base-field boundary shallow family of size 15775933:*

- (i) *affine codeword-span ranks at most five are impossible;*
- (ii) *every rank-six whole affine chart has size at most 908116, so rank six is impossible;*
- (iii) *at rank seven, every union size $g \leq 72427$ and every $g \geq 354999$ is paid; every survivor lies in*

$$\boxed{72428 \leq g \leq 354998},$$

an interval of 282571 integers;

- (iv) *ranks at least eight remain open.*

Source and status. Scott Hughes, [Hug26e, Hug26f].

Proposition 7.8 (Quotient-shell counterexample floors). *On the pinned $c = 2048$, $(u, v) = (0, 1)$ quotient profile:*

- (i) *a T_{64} class-swap family refutes the proposed $Q-3+4$ rooted-shell statement;*
- (ii) *an anchor has 1233 distinct depth-32-prefix neighbors at deficiency 192, including 1225 triple- T_{64} swaps and eight genuinely T_{16} -mixed neighbors;*
- (iii) *every coefficient-four uniform intercept at that deficiency must satisfy $1233 \leq b \leq 5191$ if it is to remain arithmetically viable.*

Source and status. Holm Buar, [Bua26c, Bua26d]. These are exact support-level counterexamples, not received-line or slope lower bounds.

Theorem 7.9 (Scalar descent and post-Johnson contract). *At agreement 1116023, the deployed quartic-field list inequality*

$$B_{\mathbb{F}_{p^4}}(1116023) \leq 16777215$$

is equivalent to the corresponding prime-field inequality. Separately, a CA numerator at most 16777214 for $\text{RS}(D, 2^{20})$ would imply list size at most 16777215 for $\text{RS}(D, 2^{20} + 1)$ and hence for the original dimension- 2^{20} code.

Source and status. Scalar descent: Danny, [Dan26c]. Exact CA-to-list arithmetic: Holm Buar, [Bua26e], using Theorem 5.3 of [ABF26]. The required CA premise remains open.

8. CHEBYSHEV, ANALYTIC, AND FORMAL PACKETS

Theorem 8.1 (D_{128} Chebyshev package). *Over $\mathbb{F}_{2^{31}-1}$ on the normalized 128-point root set of T_{128} :*

- (i) *the first three power sums are injective on block-free seven-subsets;*
- (ii) *every fixed- e_1 affine coefficient line containing at least three canonical monic quartics is an opposite-pair carrier*

$$(Y^2 - sY + p)(Y^2 + q),$$

and there are exactly 8128 heavy carrier lines among 10667968 canonical quartets;

- (iii) *if $d = 2^s \geq 4$, $n = M/d$, and a monic degree- d polynomial has r distinct simple fibers completely split over the roots of T_M , then $3r \geq 2(n + 1)$ forces a quadratic right factor. Without such a factor, there are at most $\lfloor (2n + 1)/3 \rfloor$ complete split fibers; at $n = 32$ the cap is 21 for $d = 8, 16, 32$.*

Source and status. Manuel E. Rey-Álvarez Zafiria, [Rey26a, Rey26c, Rey26b].

Proposition 8.2 (Typed pair-resource occupancy). *The active pair resource splits exactly as*

$$15841 = 12285 + 3556$$

into base and partial-block pairs. A universal two-price occupancy inequality holds, with primal-dual optimal prices for typed codegrees $(21, 2)$. Under the additional global codegree hypotheses printed in the source, the refined Mersenne transfer residual 1761515 is bounded by 1756055, with slack 5460.

Source and status. Manuel E. Rey-Álvarez Zafiria, [Rey26d]. The resource theorem is unconditional; the displayed Mersenne transfer application is conditional on the stated global codegree bounds.

Proposition 8.3 (Exact dense-shell reductions). *The dense-shell programme unconditionally provides:*

- (i) *an exact character-factorization formula for every support-class sum;*
- (ii) *the positive one-coordinate atom inequality on the full stated interval;*
- (iii) *the corrected parity-sensitive positive-charge identity*

$$\omega(\text{class}(U)) = \frac{M_U + \Sigma_U}{2} = W_U + \mathbf{1}_{\{s_U=+1\}}|\Sigma_U|;$$

- (iv) *a certified insertion dynamic program with an analytic truncation bar;*
- (v) *counterexamples to the parity-free charge identity and to two overstrong decorated-cone assertions.*

The all-depth MASTER and INV-TAIL claims remain conditional or computed-grid.

Source and status. [Dense26].

Proposition 8.4 (Singleton-MASTER reductions). *For the all-position single-insertion operator,*

$$D_{n,q} = T^{q-1}SB_{n-q}.$$

First-insertion source positivity follows from ordinary MASTER; a normalized derivative/Helmholtz inequality is sufficient for all-position monotonicity; the terminal coordinate is positive and satisfies the exact recurrence in the source. The proposed cap-six source envelope is false, and every shifted-Jacobi source condition $(K - \lambda I)D \geq 0$ with $\lambda > 1/10$ is false.

Source and status. Vadim Avdeev, [Avd26]. The corrected source envelope and its preservation under T remain open.

Remark 8.5 (Formalized kernels and permanent regressions). The repository also contains theorem-level or audit-level formal packages for complete-fiber subset packing, the ECFFT rational floor and one-step cap, the repaired φ -fiber interface, and the $c = 1$ first-grid cap chain. Exact counterexamples rule out a universal strict-crossing demand, selected-support/unique-mate top-seam shortcuts, and the incorrect support universe $\binom{|\mathbb{B}|}{m}$ in place of $\binom{|D|}{m}$. The promotion and nonpromotion map is in [PR26]; no stale, source-unbound, or computed-grid certificate is used as a theorem here.

9. INTERPLAY AND THE BEST CURRENT SUBMISSION PACKAGE

TABLE 5. Logical interplay of the strongest result families.

result family	what it proves	how it enters the Prize submission
exact staircase and Proth rows	complete thresholds for explicit smooth rows	headline partial-Prize theorem
$\mathbb{F}_{17^{32}}$ tangent threshold	exact adjacent threshold on a smooth subgroup	second complete solved row
corridor safe edges	strongest unconditional large-field safe radii at rates $1/4, 1/8, 1/16$	safe side of the best current intervals
universal cap and identity floors	uniform and deployed failure radii	unsafe side of the interval tables
high-agreement ledger	exact line, CA, projective-slope, curve, and list numerators	protocol-facing coding compiler
Koala tangent and localization packets	one banked atom plus rational/ray restrictions	inputs toward spread, owner, and exception routing
Mersenne source/rank/Chebyshev packets	low cell payment, exact residual reductions, finite rank closures	inputs toward the terminal spread/owner theorem
Delsarte barrier and critical cores	rules out ordinary LP closure and forces abundant structure	directs the next theorem to three-point or algebraic geometry

For the two exact deployed MCA adjacent rows, the authoritative finite architecture remains

$$\boxed{\text{spread routing } S + \text{large-owner image } A + \text{exception routing } E.}$$

The unconditional packets in this paper feed those three inputs but do not prove them completely. The remaining finite proof requires classification of every positive-dimensional nonproper correction component, the row-sharp large-owner challenge-image bound, chronology-correct routing of denominator roots and all other exceptional branches, and one literal exact replay with no unresolved cells. The detailed contract is stated in [ChoComp26].

The cleanest external submission package is therefore:

- (1) this results paper, centered on complete exact rows, corridor safe edges, universal unsafe caps, and the exact line/curve/list ledger;

- (2) a deployed-frontier companion containing the KoalaBear and Mersenne structural packets and the $S+A+E$ endgame;
- (3) a reproducibility dossier containing literal row manifests, primality certificates, source pins, exact replays, and formal correspondence files.

10. CONCLUSION

The repository now contains several complete solutions of genuine Proximity Prize subfamilies, including four maximal-dimension smooth prime-field thresholds and the exact $\mathbb{F}_{1732}$ smooth row. It also contains the strongest currently audited large-field safe edges at rates $1/4$, $1/8$, and $1/16$, universal near-capacity failure caps, exact deployed unsafe floors, and a row-independent line/curve/list coding ledger. The KoalaBear and Mersenne programmes have advanced from undifferentiated residuals to banked atoms, precise algebraic localizations, finite rank closures, exact counterexample floors, and a forced critical-core dichotomy. The adjacent deployed safe rows are still open and are not claimed here. Every result in this paper is stated without reproducing its proof and points to the paper, note, or certificate packet where the proof and exact arithmetic are available.

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